



ROUND-TRIP RATE FOR RRP

- Hence the round trip rate satisfies:

$$\tau_{\text{RPT}} = \frac{1}{2N + 2 \sum_{n=1}^N \frac{r_n}{1 - r_n}} \leq \frac{1}{2N}$$

► RPT is very sensitive to both the annealing schedule number of chains N

► **The first term** is expected round trip time for a simple random walk $2(N+1)N$

The second term is the additional expected time due to rejection

► Moreover as $N \rightarrow \infty$ we have $\tau_{\text{RPT}} \rightarrow 0$ independent the schedule.

► When N is small rejection dominates and N is large diffusivity dominates



Difficult to turn obsolete into hardware

► **Theorem:** Let $r_n = s_{n,-}$ be the rejection probability for the swap between chains $n-1$ and n

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$$\mathbb{E}_{\text{RPT}}[\Delta T] = 2N(N+1) + 2(N+1) \sum_{n=1}^N \frac{r_n}{1-r_n}$$

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- ▶ **The second term** is the additional expected time due to rejections
- ▶ RPT is very sensitive to both the annealing schedule number of chains N
 - ▶ One high rejection rate will make performance collapse
 - ▶ When N is small rejection dominate and N is large diffusivity dominates
- ▶ Difficult to tune or scale to modern hardware

ROUND-TRIP RATE FOR NRPT